

SJ(govt) polytechnic ,Bangalore

COMPUTER SCIENCE AND ENGINEERING

Title : CIE-2

Testheapstack software solutions pvt. Ltd.

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HEAD OF THE DEPT: HARITHA S M

Agenda of the presentation:

1. Intern's ability to apply the skill and technical knowledge on OJT-1
2. Intern's performance on assigned tasks and project
3. Extent of Intern's ability to add value to the organization through internship
4. Document a Use case on a task where he/she is working as intern

- *Project Overview – Urban Cost Prediction

- *Job Roles and Responsibilities

- *Technical Skills Applied

- *Use Case Description

- *Value Added and Learning

- *Conclusion

Project Overview – Urban Cost Prediction

Urban Cost Prediction is a machine learning project that predicts the cost of living in urban areas.

The system analyzes different factors such as location, housing prices, transportation, and daily expenses.

It uses data analysis and ML algorithms to estimate the expected cost.

This helps users plan their budget and make better financial decisions before moving to a city.

The model processes the data and provides accurate cost predictions based on input features.

USING MACHINE LEARNING TECHNIQUES



OJT-1 : **Technical Skills Applied**

- *Python Programming for developing the machine learning model

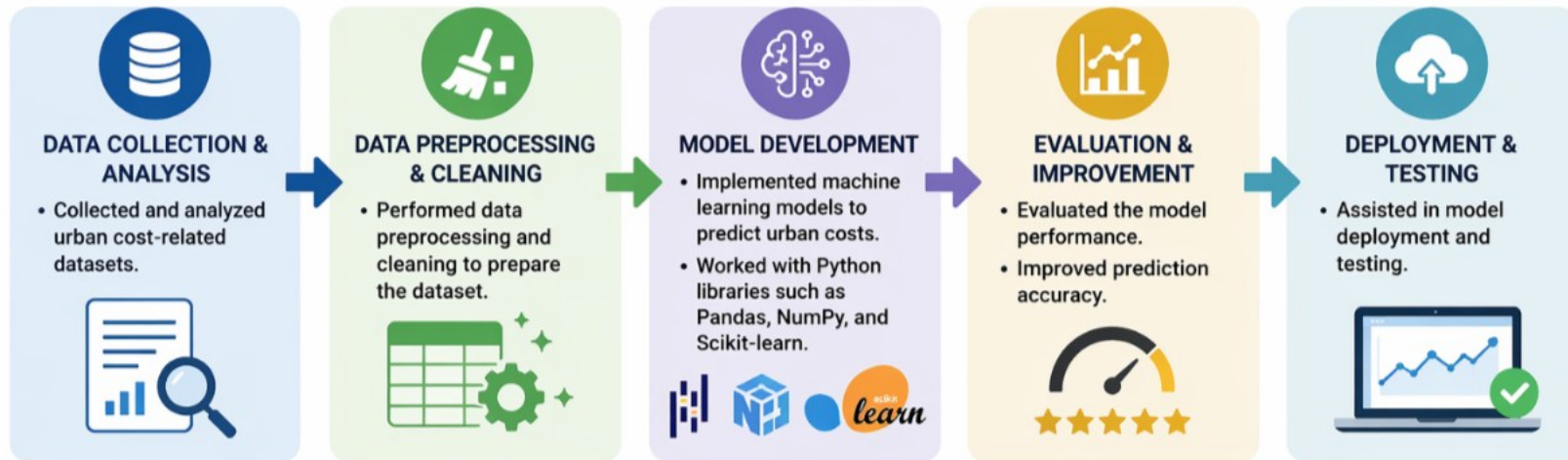
- *Pandas & NumPy for data analysis and preprocessing

- *Scikit-learn for implementing machine learning algorithms
Data Visualization using Matplotlib / Seaborn

- *Model Training and Evaluation to improve prediction accuracy
Basic Model Deployment

Urban Cost Prediction Project

Data-Driven Insights for Smarter Cities



Job Roles and Responsibilities

{Intern's performance on assigned tasks and project }

- *Collected and analyzed urban cost-related datasets.
- *Performed data preprocessing and cleaning to prepare the dataset.
- *Implemented machine learning models to predict urban costs.
- *Worked with Python libraries such as Pandas, NumPy, and Scikit-learn. Evaluated the model performance and improved prediction accuracy.
- *Assisted in model deployment and testing.

Project – Urban Cost Prediction

Project Details

Area (sq km)
18.50

Population Density
1900

Infrastructure Type
Mixed

Land Cost per sq km
310000

Construction Cost Index
1.10

Distance to City Center (km)
17.00

Existing Infrastructure Score (0.2-1.0)
0.50

Urban Planning Cost Prediction

This application predicts the total cost of urban planning projects based on various factors. Enter the project details below to get an estimated cost prediction.

Cost Prediction Result

Predicted Cost [?]
\$374,006.09 ...

Predicted Cost (USD) [?]
\$374,006,092,...

Cost per sq km [?]
\$20,216.55 Mi...

Input Summary

	Value
Area (sq km)	18.5
Population Density	2350
Land Cost per sq km	310000
Construction Cost Index	1.2
Distance to City Center (km)	15
Existing Infrastructure Score	0.5

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Use Case Description

This project predicts the cost of living in urban areas using machine learning. It helps users estimate expenses and plan their budget before moving to a city.

Steps Involved

User Input: User enters details like location, housing cost, transportation, and daily expenses.

Data Processing: The system processes and analyzes the input data.

Model Prediction: The machine learning model predicts the estimated urban cost.

Result Display: The system shows the predicted cost to the user.

Decision Making: Users can use this prediction to plan their budget and expenses.

USE CASE DESCRIPTION

A use case describes how a user (actor) interacts with the system to achieve a goal.

PROJECT: URBAN PLANNING COST PREDICTION



USE CASE NAME:

Predict Urban Planning Project Cost



ACTOR (USER):

Urban Planner / Government Official /
Project Consultant



GOAL

To predict the estimated cost of urban planning and development projects accurately using historical data and machine learning.



PRECONDITION

- The user must be registered and logged in.
- Historical project data must be available in the system.



MAIN FLOW (PROCESS)

01



User inputs project details (e.g., area, location, infrastructure type, materials).



02



System preprocesses the data and extracts relevant features.



03



Machine learning model analyzes the data and predicts the cost.



04



System displays the predicted cost along with key factors.



05



User reviews the results and adjusts inputs if needed.



06



User downloads or saves the cost report for reference.



ALTERNATE FLOW

- If input data is incomplete or invalid, the system shows an error message.
- If the model confidence is low, the system recommends adding more details.



POSTCONDITION

The system provides a cost estimate report, helping the user make informed budgeting and planning decisions.



BENEFIT

- Saves time and effort in cost estimation.
- Improves budgeting accuracy.
- Supports data-driven urban planning decisions.

Value Added and Learning

Value Added:

- *Gained practical experience in Machine Learning and Data Analysis.
- *Improved Python programming skills.
- *Learned to work with real-world datasets.

Learning Outcomes:

- *Understood the process of data preprocessing and feature selection.
- *Learned how to build and train machine learning models.
- *Gained knowledge of model evaluation and prediction techniques.
- *Developed analytical and problem-solving skills.
- *Learned the basic steps of deploying a machine learning model.

Conclusion:

- During this internship, I worked on the Urban Cost Prediction project, where I applied machine learning techniques to analyze urban cost data and predict living expenses.
- The project involved data preprocessing, model training, and prediction analysis.
- This project helped me understand the real-world application of machine learning and data analysis.
- The internship provided valuable practical experience and improved my technical and analytical skills.